

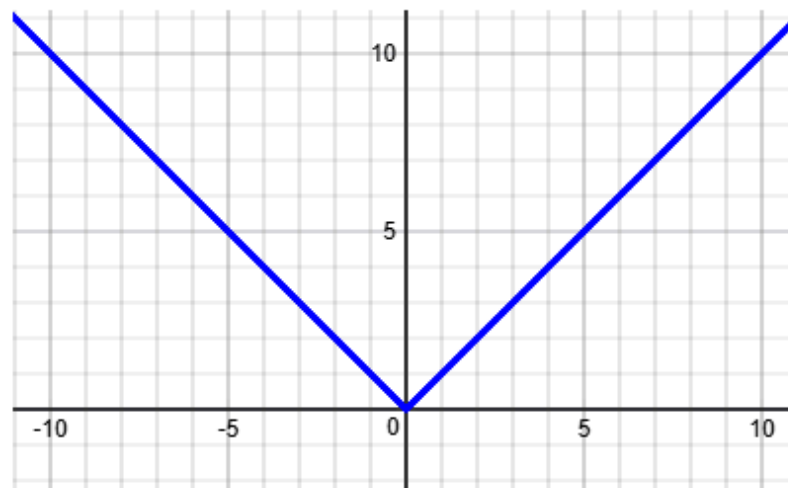
Calculus I Lesson 15

- 1) If $x \rightarrow 0$ from the left, is x positive or negative?
- 2) If $x \rightarrow 0$ from the right, is x positive or negative?
- 3) If $x \rightarrow 2$ from the left, is $x - 2$ positive or negative?
- 4) If $x \rightarrow 2$ from the right, is $x - 2$ positive or negative?
- 5) If $x \rightarrow 2$ from the left, is $2x - 4$ positive or negative?
- 6) If $x \rightarrow 2$ from the right, is $2x - 4$ positive or negative?

Alternative definition of a function $f(x)$ at c : $f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$

$$7) f(x) = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}.$$

$$\text{Find } f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0}.$$



$$\text{a) } \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \underline{\hspace{2cm}} ?$$

$$\text{Hint: } \frac{f(x) - f(0)}{x - 0} = \frac{-x - 0}{x - 0}$$

Note: $x \rightarrow 0$ from the left; so x is negative and $f(x) = -x$.

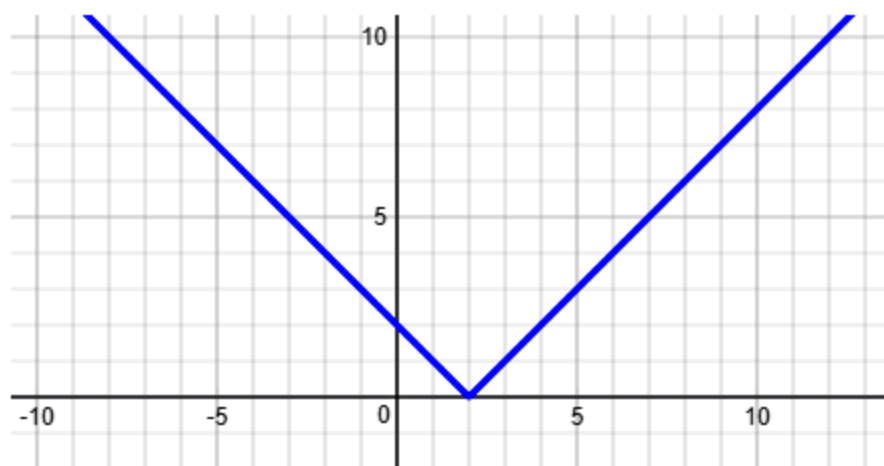
b) $\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \underline{\quad ? \quad}$ Hint: $\frac{f(x) - f(0)}{x - 0} = \frac{x - 0}{x - 0}$

Note: $x \rightarrow 0$ from the right; so x is positive and $f(x) = x$.

c) $f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \underline{\quad ? \quad}$

d) Is $f(x)$ differentiable at $x = 0$?

$$8) f(x) = |x - 2| = \begin{cases} x - 2 & \text{if } x - 2 \geq 0 \\ -(x - 2) & \text{if } x - 2 < 0 \end{cases}$$



Find $f'(2) = \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}$.

a) $\lim_{x \rightarrow 2^-} \frac{f(x) - f(2)}{x - 2} = \underline{\hspace{2cm}} ?$ Hint: $\frac{f(x) - f(2)}{x - 2} = \frac{-(x - 2) - 0}{x - 2}$

Note: $x \rightarrow 2$ from the left; so $x - 2$ is negative and $f(x) = -(x - 2)$

b) $\lim_{x \rightarrow 2^+} \frac{f(x) - f(2)}{x - 2} = \underline{\hspace{2cm}} ?$ Hint: $\frac{f(x) - f(2)}{x - 2} = \frac{x - 2 - 0}{x - 2}$

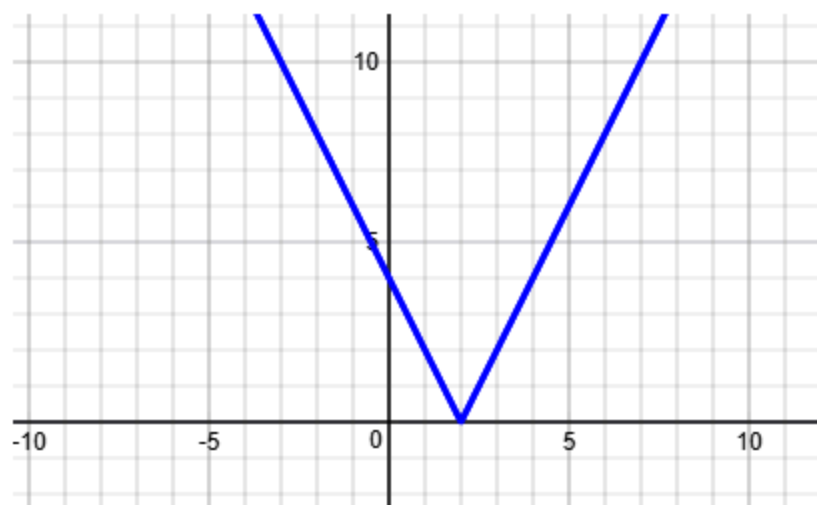
Note: $x \rightarrow 2$ from the right; so $x - 2$ is positive and $f(x) = x - 2$

c) $f'(2) = \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} = \underline{\quad ? \quad}$

d) Is $f(x)$ differentiable at $x = 2$?

$$9) f(x) = |2x - 4| = \begin{cases} 2x - 4 & \text{if } 2x - 4 \geq 0 \\ -(2x - 4) & \text{if } 2x - 4 < 0 \end{cases}$$

$$\text{Find } f'(2) = \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}.$$



$$\text{a) } \lim_{x \rightarrow 2^-} \frac{f(x) - f(2)}{x - 2} = \underline{\hspace{2cm}} \quad \text{Hint: } \lim_{x \rightarrow 2^-} \frac{f(x) - f(2)}{x - 2} = \frac{-(2x - 4) - 0}{x - 2} = \frac{-2(x - 2)}{x - 2}$$

Note: $x \rightarrow 2$ from the left; so $2x - 4$ is negative and $f(x) = -(2x - 4)$

$$\text{b) } \lim_{x \rightarrow 2^+} \frac{f(x) - f(2)}{x - 2} = \underline{\hspace{2cm}} \quad \text{Hint: } \lim_{x \rightarrow 2^+} \frac{f(x) - f(2)}{x - 2} = \frac{2x - 4 - 0}{x - 2} = \frac{2(x - 2)}{x - 2}$$

Note: $x \rightarrow 2$ from the right; so $2x - 4$ is positive and $f(x) = 2x - 4$

c) $f'(2) = \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} = \underline{\hspace{1cm}} ?$

d) Is $f(x)$ differentiable at $x = 2$?

